BridgeU: A Bilingual Community Platform for International Students in Thailand

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***Abstract*—International students in Thailand face significant barriers in accessing local news, building social connections, and adapting to an unfamiliar environment. This paper presents BridgeU, a bilingual (Chinese/English) web-based community platform designed to address these challenges. BridgeU provides three integrated features: a Daily Briefing system that automatically aggregates Thai news via Google News RSS and translates it into Chinese and English using Qwen AI; a Community Page that enables peer-to-peer knowledge sharing with AI-powered moderation and translation; and an Authentication and Profile system with email and SMS verification. The system is built using Vue.js 3 on the frontend and Spring Boot 3 on the backend, with MySQL 8 as the primary database and JWT-based stateless authentication. A competitive analysis against Facebook Groups, Xiaohongshu, and Reddit confirms that BridgeU uniquely combines bilingual AI news aggregation with a structured, moderated community space. Testing covered 25 unit test suites and 21 system test cases across all features, confirming that all core functional requirements were satisfied.**

Keywords—international students, community platform, news aggregation, bilingual interface, Vue.js, Spring Boot, JWT authentication, Qwen AI, cross-cultural adaptation

I. INTRODUCTION

The globalization of higher education has driven rapid growth in international student mobility. According to UNESCO, the number of internationally mobile students exceeded 6 million by 2022, with institutions in Southeast Asia, including Thailand, receiving a growing share [1]. Chinese students represent one of the largest groups of internationally mobile students globally, and their presence at Thai universities has increased substantially in recent years.

International students frequently encounter compounded challenges upon arrival in a host country. Research by Hofhuis et al. [2] demonstrated that social network site (SNS) use significantly affects the wellbeing of international students through acculturation and adaptation processes. The study found that interactions with host-country contacts on SNS were positively associated with sociocultural adaptation, while isolation from local information sources remained a persistent barrier. Similarly, Zhou and Yin [3] conducted an integrative review of research on international students' social media use, concluding that platform affordances play a critical role in determining whether technology facilitates or impedes cross-cultural adjustment.

In Thailand, international students face a specific challenge: the majority of local news, public announcements, and community information is published exclusively in Thai. Existing general-purpose tools such as Facebook Groups, Xiaohongshu, and Reddit do not fully address this need, as analyzed in Section II-B. Furthermore, no dedicated platform exists to connect Chinese-speaking international students in Thailand or provide a unified community space tailored to their needs.

To address these gaps, this paper proposes and evaluates BridgeU, a bilingual web-based community platform. BridgeU is designed around three core features: (1) the Daily Briefing System automatically collects Thai news via Google News RSS and translates summaries into Chinese and English using Qwen AI; (2) the Community Page provides an AI-moderated forum for likes, comments, and follow relationships; and (3) the Authentication and Profile System supports registration via email or SMS verification and JWT-based session management. This paper describes the design rationale, competitive positioning, system architecture, feature implementation, and testing outcomes.

II. RELATED WORK

A. Challenges Faced by International Students

A growing body of literature documents the multi-dimensional difficulties experienced by international students. Gautam et al. [4] identified language barriers, cultural distance, loss of social support networks, and academic adjustment as primary stressors. Dong et al. [5] found that social media usage significantly affects social capital and learning engagement among international students in the United States, highlighting the dual role of digital platforms as both connectors and potential sources of isolation. The British Council [6] further reported strong demand among international students for private, topic-specific digital communities where they can discuss practical matters such as visa policies, housing, and affordability. These findings directly motivate the design of BridgeU as a specialized, structured platform rather than a repurposed general-purpose social network.

B. Competitive Analysis of Existing Platforms

Several widely used platforms are currently adopted by international students in Thailand as informal information and community channels. A comparative analysis of three key platforms—Facebook Groups, Xiaohongshu (Little Red Book), and Reddit—reveals important limitations that BridgeU is designed to overcome.

1) Facebook Groups:

Facebook Groups are currently one of the most widely used mutual-aid tools among international students worldwide [15]. Their advantages include a large existing *user base and broad reach. However, Facebook Groups have a high language threshold: content is predominantly in Thai and English, making them difficult for Chinese students to use. Information retrieval is extremely poor, with no structured tags, so rental listings, job posts, and casual chat are all mixed together, making targeted search effectively impossible.*

2) Xiaohongshu (Little Red Book):

Xiaohongshu is the most popular lifestyle sharing platform among Chinese international students [16]. It offers rich content, algorithm-based recommendations, and an aesthetically pleasing interface. However, it suffers from a serious "tourist perspective" bias: content is overly focused on travel, food, and check-ins, and lacks in-depth, practical accumulation for long-term residents. Its closed Chinese ecosystem means it is impossible to reach local Thais, landlords, or international students from other language backgrounds.

3) Reddit:

Reddit is one of the world's largest online forums and hosts numerous study-abroad-related subreddits (e.g., r/Thailand, r/studyabroad) [17]. It offers structured topic-based categorization and an environment where users feel comfortable asking genuine questions under pseudonymity. However, its interface is unfriendly to non-technical users, and it operates as a pure English environment with no multilingual translation support, effectively excluding most Chinese students and local Thai community members.

TABLE I. COMPETITIVE FEATURE COMPARISON

Feature	**Facebook Groups**	**Xiao-hongshu**	**Reddit**	**BridgeU**
Web-based Access	✓	✓	✓	✓
Bilingual (CN/EN) Interface				✓
AI News Aggregation (RSS)				✓
AI Auto-Translation	✓	✓		✓
AI Content Moderation				✓
Structured Post Tags			✓	✓
Community Forum / Post Feed	✓	✓	✓	✓
Thailand-Focused Content				✓

As shown in Table I, none of the existing platforms combines bilingual AI news aggregation, structured post tags, AI-powered content moderation, and Thailand-focused community features in a single free-to-use web platform. BridgeU is designed to fill this gap by integrating all of these capabilities.

C. News Aggregation and Automatic Translation

RSS (Really Simple Syndication) has been established as a practical and standardized mechanism for news aggregation. Kasioumis et al. [7] proposed a distributed RSS platform demonstrating the scalability of RSS-based approaches. Atlas [8] showed that combining RSS feeds with third-party news APIs significantly improves content coverage. The integration of large language models into translation pipelines has become increasingly feasible. Yuen and Schlote [9] reviewed AI applications in language learning and noted that AI-powered tools can bridge cultural and linguistic gaps effectively, provided appropriate quality controls are in place. BridgeU applies this principle by using Qwen AI (Alibaba Cloud) to generate Chinese and English summaries at ingestion time, so translated content is always immediately available to users.

D. Web Application Architecture

Wang et al. [10] demonstrated the effectiveness of Vue.js and Spring Boot for building scalable online education platforms, noting that Vue.js's reactivity system and component-based model reduce frontend complexity while Spring Boot's auto-configuration accelerates backend development. Kaya and Ozdemir [11] further evaluated Spring Boot-based REST API implementations under the OWASP security framework, identifying JWT authentication as the recommended stateless session mechanism. BridgeU adopts this architecture with Vue.js 3 on the frontend and Spring Boot 3 on the backend, connected via RESTful APIs over HTTPS. MySQL 8 is used as the primary relational database, and Docker provides containerized deployment.

E. Authentication and Security

JSON Web Tokens have become the de facto standard for stateless authentication in modern web applications. Bucko et al. [12] demonstrated that JWT-based authentication can be further strengthened by incorporating user behavior history for anomaly detection. The OWASP Cheat Sheet for JWT [13] prescribes best practices including HTTPS for all token transmissions, short expiry windows, and server-side signature validation. BridgeU implements these practices: passwords are hashed using BCrypt with salt before storage, tokens are signed with HMAC-SHA256, and all protected endpoints validate the token on every request.

F. Community Platforms and AI-Assisted Moderation

Hofhuis et al. [2] showed that interactions within peer networks on SNS platforms are associated with improved adaptation outcomes for international students. Roczey and Szenasi [14] proposed an automated moderation helper system using AI-based text classification, demonstrating that machine learning classifiers can reliably flag rule-violating content with high precision. BridgeU applies Qwen AI for both automatic translation of user-generated content and content security review before publication, maintaining community quality without requiring immediate manual review for every submission.

III. PLATFORM DESIGN AND IMPLEMENTATION

A. System Overview

BridgeU is designed as a web-based platform targeting Chinese-speaking international students at Thai universities. The frontend is implemented with Vue.js 2 and Element UI, using Axios for HTTP communication between the frontend and backend. The backend is built with Spring Boot, exposing RESTful APIs and using MyBatis as the persistence framework for database access. MySQL 8 serves as the primary relational database for storing structured user and post data. Elasticsearch is integrated to provide high-performance semantic search capabilities. The system also connects to the Aliyun Qwen API to perform natural language processing tasks such as summarization and translation. The overall system architecture is illustrated in Fig. 1.

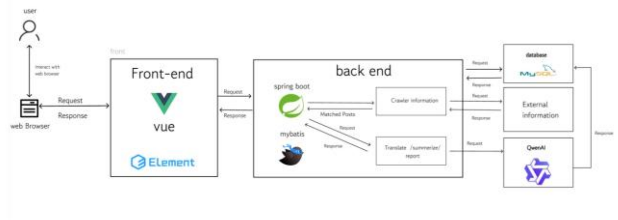

Fig. 1. System Architecture

B. Feature 1: Daily Briefing System

The Daily Briefing System addresses the most critical information barrier: the inaccessibility of Thai-language local news for non-Thai-speaking international students. At 08:00 daily, the system crawls Google News RSS feeds. For each item, the system calls the Qwen AI API to generate Chinese and English versions of the title and summary, which are stored persistently in the database. This batch-preprocessing approach eliminates real-time API latency during user browsing [9].

The frontend renders a paginated list of 10 news cards per page, each showing the translated title, AI-generated summary, source label, and publication date in the user's current language. Users can search by keyword across both Chinese and English translation fields simultaneously, filter by publication date range, and toggle between Chinese and English interfaces dynamically. The backend's Data Transfer Object (DTO) conversion suppresses all Thai original text from user-facing responses. Custom JPQL repository queries support keyword search (findByKey word),date-range filtering (findByPublishDateBetween Ordered), and combined filters, all returning results ordered by publication date descending. The feature interface is shown in Fig. 2.

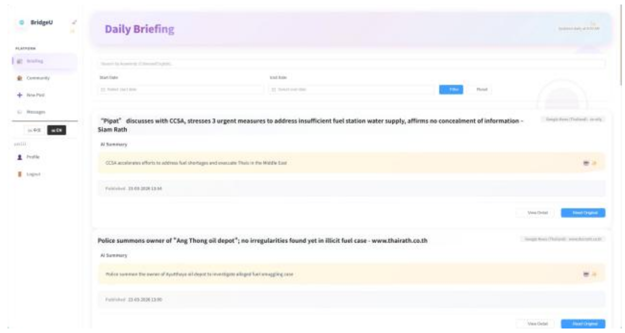

Fig. 2. Daily Briefing System interface

C. Feature 2: Community Page

The Community Page enables peer-to-peer knowledge sharing among international students across five predefined categories: Study, Housing, Travel, Part-time Job, and Life Services. When a user submits a post in Chinese or English, the system asynchronously calls Qwen AI to generate the other language version and conduct a content security review. The original content is immediately saved and the user receives a response, while AI processing runs in the background [14]. Only after Qwen AI confirms the content passes moderation checks does the post appear in the community feed.

Within each post, users may add comments, toggle a like reaction, and generate an AI-summarized paragraph of all comments—useful for long discussion threads. The follow relationship is asymmetric: mutual follows create a "Friends" relationship enabling unrestricted private messaging, while one-way follows are limited to a single initial message to reduce unsolicited contact. Users can report posts or comments by selecting predefined reasons (Spam, Fraud or Scam, Illegal Service Promotion, Abusive Language, Other), after which Qwen AI reviews the reported content and provides a judgment. The Community Page also supports keyword search across post titles and body texts in currently languages. The Community Page feed, New Post form, and private messaging interface are shown in Fig. 3, Fig. 4, and Fig. 5 respectively.

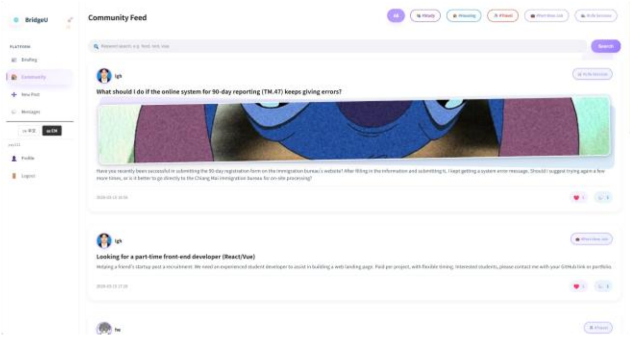

Fig. 3. Community Page feed with category filters and post cards

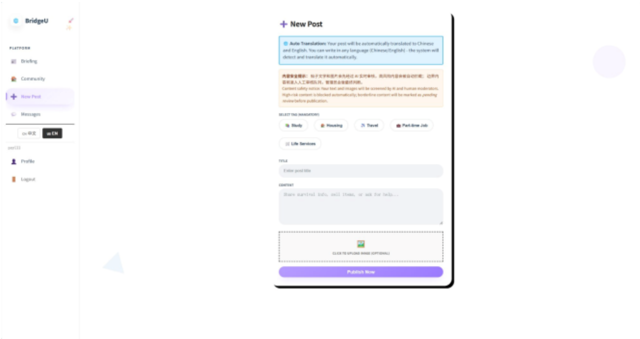

Fig. 4. Create New Post interface

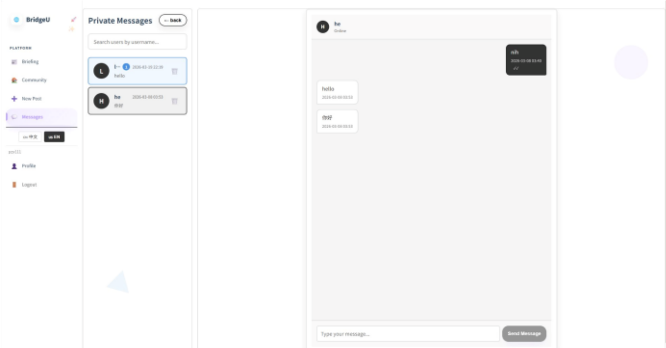

Fig. 5. Private messaging interface showing conversation threads

D. Feature 3: Authentication and Profile System

The Authentication and Profile System provides secure user registration, login, and profile management. Registration supports two verification pathways: email (via server-side SMTP verification code) and phone (via Firebase SMS). Both pathways require a 6-digit verification code before the registration form is unlocked. Passwords are validated to contain both uppercase and lowercase letters and are stored as BCrypt hashes [13]. Display names are restricted to English characters to prevent encoding conflicts. If the user forgets the password, the option to forget the password is provided on the login page, and the user can send a verification code to reset the password through the phone number or email.

Upon successful login, Spring Security generates a JWT signed with HMAC-SHA256, containing the user's ID, username, and token expiry. The client stores the token in sessionStorage and attaches it as a Bearer token in the Authorization header of all authenticated requests. The Profile page displays avatar, display name, username, and social statistics (posts, followers, mutual follows). Users can edit their display name, avatar URL, and preferred language; changing the username requires re-authentication. Additional profile views include My Community Posts (with Qwen AI moderation status: published, pending, or rejected), My Reports (showing AI review judgments for submitted reports), and Followers/Friends lists with direct navigation to user profiles or the messaging interface when click the user's avatar. The feature interface is shown in Fig.6.

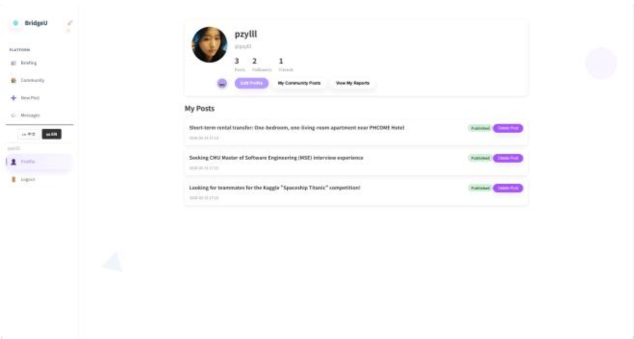

Fig. 6. User profile page

IV. Testing

A. Test Strategy

Testing followed a structured three-level strategy: unit testing, system testing, and integration testing. Unit tests targeted individual frontend Vue.js component methods and backend Spring Boot controller, service, and repository methods. System tests verified end-to-end workflows through the fully integrated system. A total of 25 unit test suites (UTC-01 through UTC-25) and 21 system test cases (STC-01 through STC-21) were defined. The pass/fail criterion required an exact match between actual results and the expected results specified in the Test Plan document.

B. Unit Testing Summary

Feature 1 (Daily Briefing) contributed 13 unit test suites covering frontend methods—fetchDailyBriefing, fetchNewsDetail, handleSearch, applyFilters, resetFilters, handlePageChange, normalizeDateValue, and formatDate—and backend methods—convertToDTO,getDailyBriefing,getNewsDetail, findByKeyword, and findByPublishDateBetweenOrdered.

Feature 2 (Community Page) contributed 8 suites covering fetchPosts, fetchPostDetail, createPost, addComment, getCommentSummary, deleteComment, toggleLike, and searchAll.

Feature 3 (Authentication) contributed 4 suites covering registerWithVerification, login, Auth API wrapper functions, and user follow API wrapper functions. All unit test cases passed upon final execution.

C. System Testing Summary

System tests were grouped into 21 scenarios spanning all implemented features, as shown in Table II. Each scenario specifies a test script with numbered steps, test cases with descriptions, inputs, and expected results, and supporting UI screenshots captured from the deployed system. All 21 system test scenarios ultimately passed; several defects identified during initial execution were corrected and verified in retesting.

TABLE II. System Test Case Summary

STC ID	Description	Feature	Result
STC-01	View Daily Briefing list with default parameters	Daily Briefing	Pass
STC-02	Filter news by publication date range	Daily Briefing	Pass
STC-03	Search Daily Briefing by keyword	Daily Briefing	Pass
STC-04	Switch interface language (EN/ZH)	Daily Briefing	Pass
STC-05	View news item detail page	Daily Briefing	Pass
STC-06	Open original news article link	Daily Briefing	Pass
STC-07	View community feed list with tag filters	Community	Pass
STC-08	View post detail and interact (like, comment, follow)	Community	Pass
STC-09	Create new post with field validation	Community	Pass
STC-10	Report post or comment	Community	Pass
STC-11	Manage message status and user search	Community	Pass
STC-12	Login with valid and invalid credentials	Auth	Pass
STC-13	Registration with email verification	Auth	Pass
STC-14	Registration with phone (Firebase SMS)	Auth	Pass
STC-15	Logout	Auth	Pass
STC-16	Forgot password via email	Auth	Pass
STC-17	Forgot password via phone	Auth	Pass
STC-18	View and edit user profile	Auth	Pass
STC-19	View my community posts	Auth	Pass
STC-20	View my submitted reports	Auth	Pass
STC-21	View friends and followers list	Auth	Pass

V. User Evaluation

A. Evaluation Design

To assess BridgeU's usability and relevance to its target users, a two-phase evaluation was conducted combining semi-structured interviews and a structured questionnaire. The evaluation targeted international students currently enrolled at Chiang Mai University who were not involved in the system's development. A total of five participants took part in the interview phase and fifteen participants completed the questionnaire, yielding qualitative depth alongside quantitative breadth. The combined approach follows the mixed-methods evaluation strategy recommended by Creswell and Plano Clark [20], wherein qualitative findings are used to interpret and contextualize quantitative results.

B. Semi-Structured Interview

Five international students, all enrolled in undergraduate programs at CAMT, participated in individual semi-structured interviews lasting 20–30 minutes each. Interviews were conducted in English or Chinese according to participant preference. The interview guide covered four themes: current information-access habits, first impressions and navigability of BridgeU, perceived usefulness of each feature, and suggestions for improvement. Thematic analysis [21] was applied to the transcripts, and three recurring themes emerged across participants.

Theme 1 — Daily Briefing as a primary news channel. All five participants stated they currently rely on Xiaohongshu or WeChat moments for Thai news, and found the Daily Briefing System more trustworthy because it sources directly from established Thai media outlets. Four participants explicitly praised the bilingual toggle, noting that switching between English and Chinese without reloading the page felt “natural” and “smooth.” One participant noted: “I did not know there were Thai government announcements like this. Having them in Chinese is very helpful.”

Theme 2 — Requests for richer social features and mobile access. Four participants requested a native mobile application, citing that most of their social media browsing is done on phones. Two participants suggested adding push notifications for new posts in followed tags. One participant raised a concern that the one-message limit for one-way follows might discourage initial contact, suggesting an opt-in icebreaker feature instead. These suggestions are documented as future work in Section VI.

C. Questionnaire

Following the interviews, a structured questionnaire was distributed to fifteen international students at CAMT via a shared Google Form link. Participants were asked to explore BridgeU independently for at least 20 minutes before responding. The questionnaire comprised twelve Likert-scale items (1 = Strongly Disagree, 5 = Strongly Agree) covering usability, feature usefulness, and overall satisfaction, plus one open-ended question for general comments. Table III presents the mean scores and standard deviations for all twelve items.

TABLE III. USER QUESTIONNAIRE RESULTS (N = 15, SCALE 1–5)

No	Questionnaire Item	Mean	SD
Q1	The Daily Briefing makes it easier for me to access Thai news in my preferred language.	4.53	0.64
Q2	The keyword search and date filter functions help me find relevant news quickly.	4.40	0.74
Q3	Switching between Chinese and English interfaces is smooth and intuitive.	4.60	0.51
Q4	The Community Page’s tag-based filtering helps me find posts relevant to my needs.	4.47	0.64
Q5	I feel the AI-generated news summaries are accurate and informative.	4.13	0.83
Q6	Creating and submitting a post on the Community Page is straightforward.	4.33	0.72
Q7	The AI comment summarization feature saves time when reading long discussion threads.	4.20	0.78
Q8	The registration process is clear and easy to complete.	4.27	0.70
Q9	The private messaging system is useful for communicating with other community members.	4.00	0.85
Q10	The overall interface is visually clear and easy to navigate.	4.40	0.63
Q11	BridgeU addresses information needs that existing platforms (e.g., Facebook, Xiaohongshu) do not.	4.53	0.52
Q12	I would recommend BridgeU to other international students in Thailand.	4.47	0.64
—	Overall Average	4.36	0.68

The questionnaire results show consistently positive evaluations across all three feature areas. Items related to the bilingual interface (Q3, Mean = 4.60) and the Daily Briefing’s relevance to participants’ information needs (Q1 and Q11, Means = 4.53) received the highest ratings. The lowest-rated item was Q9 (private messaging, Mean = 4.00), which is consistent with the interview finding that the one-message limit for one-way followers was perceived as a barrier to initial contact. The overall mean score of 4.36 out of 5 (SD = 0.68) indicates a high level of user satisfaction. Open-ended responses reinforced the interview themes, with ten of fifteen respondents specifically mentioning the Daily Briefing System as the feature they would use most regularly, and seven requesting a mobile application as the most desired future enhancement.

D. Evaluation Limitations

The evaluation has several limitations that should be acknowledged. First, the sample sizes (n = 5 for interviews, n = 15 for the questionnaire) are small and limited to students at a single institution; findings may not generalize to international students at other Thai universities. Second, participants used BridgeU for a constrained 20-minute session rather than in natural daily use, which may inflate novelty-driven ratings. Third, all questionnaire participants were recruited from personal networks, introducing potential selection bias. Future evaluations should employ larger, randomly sampled cohorts, longitudinal usage tracking, and validated usability scales such as the System Usability Scale (SUS) [22] to provide more robust evidence of BridgeU’s effectiveness.

VI. DISCUSSION AND CONCLUSIONS

BridgeU demonstrates that a focused, bilingual community platform can address the intersection of information barriers and social isolation faced by international students in Thailand. As shown in Table I, no existing platform—Facebook Groups [15], Xiaohongshu [16], or Reddit [17]—combines bilingual AI news aggregation, structured community tags, AI-powered content moderation, and Thailand-focused content in a single accessible web application. BridgeU uniquely provides all of these capabilities.

The Daily Briefing System successfully eliminates the language barrier for accessing local Thai news by batch-preprocessing Qwen AI translations at ingestion time. This design choice, consistent with the principle established by Yuen and Schlote [9] that AI tools are most effective when integrated seamlessly into the user workflow.

The Community Page demonstrates the viability of an AI-assisted moderation model for a niche student community. Asynchronous Qwen AI processing ensures that content review does not block user interactions, consistent with the moderation approach recommended by Roczey and Szenasi [14]. The asymmetric follow model with a one-message limit for one-way connections mitigates unsolicited contact risk. The Authentication and Profile System achieved a 100% pass rate across all test scenarios, including boundary cases for rate limiting, wrong verification codes and password strength. The JWT implementation follows OWASP best practices [13] including BCrypt password hashing, HTTPS-only token transmission, and server-side signature validation.

Several limitations warrant future attention. First, Qwen AI translation quality has not been evaluated against human translator benchmarks; future work should incorporate BLEU score evaluation. Second, the platform currently supports Chinese and English only; the architecture supports additional language pairs (Japanese, Korean, Vietnamese) with minimal structural changes. Third, the crawler is limited to publicly available sources and cannot access paywalled or private content.

In conclusion, BridgeU successfully delivers bilingual news aggregation, AI-moderated community interaction, and authenticated profile management for international students in Thailand. All 21 system test scenarios and 25 unit test suites passed. The platform demonstrates that targeted software engineering—combining RSS-based news aggregation, Qwen AI translation and moderation, Vue.js reactivity, Spring Boot REST APIs, and JWT authentication—can meaningfully reduce the information and social integration barriers documented in the literature [2], [3], [5].

Acknowledgment

The author thanks Pattama Longani, Ph.D. (Project Advisor, Software Engineering Program, College of Arts, Media and Technology, Chiang Mai University) for guidance throughout the development and documentation of BridgeU. This project was completed as part of the Bachelor of Science in Software Engineering program.

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